

Identification of Twin-Related Grain Pairs from EBSD Orientation Data

1 Problem statement

Given two crystallographically indexed grains (clusters) with average orientation matrices G_1, G_2 (mapping sample coordinates to crystal coordinates), we wish to determine whether the pair is related by a specific twinning mode, defined by its crystallographic elements K_1, K_2, η_1, η_2 (composition planes and shear directions) tabulated in a twin-system database for a set of M crystallographically equivalent variants of the mode.

The variant count M is not a free parameter: it is the size of the orbit of the twin-element tuple (K_1, η_1) (or, more generally, the full tuple $(K_1, K_2, \eta_1, \eta_2)$, depending on how the tabulation was generated) under the action of the phase's point group $\mathcal{G}$, of order N_s . By the orbit–stabiliser theorem,

$$M = \frac{N_s}{|\text{Stab}(K_1, \eta_1)|}, \quad (1)$$

so $M \leq N_s$ always, with equality only when the stabiliser of the twin-element tuple is trivial. Different twin systems within the same point group can therefore have different M ; M must be taken as tabulated for each system rather than assumed. A concrete value of M is given for the worked example in Section 6.

2 Symmetry ambiguity of measured orientations

A crystal orientation is only ever known modulo the crystal's point group $\mathcal{G}$ (order N_s): for any $S \in \mathcal{G}$, SG represents the same physical crystal. Consequently, if G_1 and G_2 are each independently resolved to *some* representative (e.g. by an unconstrained least-squares average over the grain's pixels), the two representatives carry no guarantee of being labelled on *mutually consistent* branches. The raw misorientation

$$M = G_2 G_1^T \quad (2)$$

is then only *one arbitrary member* of the physically equivalent set $\{S_2 G_2 (S_1 G_1)^T : S_1, S_2 \in \mathcal{G}\}$, and its rotation angle need not be close to the true minimum disorientation between the two grains.

Resolution. Before any twin analysis, every cluster's average orientation (and, for consistency, its constituent pixel orientations) is re-expressed on a single symmetry branch chosen so as to minimise disorientation to one shared reference grain. This is a *single* symmetry operation per cluster, applied uniformly to that cluster's average and to every one of its pixels – never resolved independently per pixel, since an independent per-pixel search was found empirically to split a single physically coherent grain's pixels across several different branches when pixel-level measurement noise straddles a branch boundary. A concrete instance of both this failure mode and its fix is given in Section 6.

3 Twin conditions used

A twinning operation can be described, for a Type I twin, as a 180° rotation about the K_1 plane normal n_1 (a reflection across K_1 is experimentally indistinguishable from this rotation for a centrosymmetric crystal, by Friedel’s law – EBSD diffraction patterns cannot distinguish a crystal from its point-inverted enantiomorph, so the effective symmetry group used throughout is the extended (Laue-class) group, improper operations included). For a Type II twin, the equivalent description is a 180° rotation about the η_1 shear direction a_1 . A third, general description (valid for compound and non-compound modes alike) is a rotation about the K_2 plane normal n_2 by the mode’s characteristic shear angle γ .

All three conditions are checked:

$$\text{K1_180: } \text{axis}(M) \parallel n_1, \quad \text{angle}(M) = 180^\circ \quad (3)$$

$$\text{eta1_180: } \text{axis}(M) \parallel a_1, \quad \text{angle}(M) = 180^\circ \quad (4)$$

$$\text{K2_shear: } \text{axis}(M) \parallel n_2, \quad \text{angle}(M) = \gamma \quad (5)$$

where $\text{axis}(M)$, $\text{angle}(M)$ are extracted from M via a sign-consistent unit quaternion, and axis comparison is sign-independent ($|\hat{n} \cdot \hat{a}|$), since a rotation axis and the tabulated crystallographic directions are each defined only up to line direction.

All three conditions are always fully evaluated for every candidate pair (never short-circuited on the first match), and every condition satisfying the tolerance is reported, not only the single best. A pair for which K1_180 and eta1_180 are *both* independently satisfied (generally via different representative rotations, see below) is a compound twin – all three descriptions coincide exactly when the mode’s K_1, K_2, η_1, η_2 are all simultaneously rational; observing K1_180, eta1_180, and K2_shear simultaneously satisfied for the same pair is accordingly the expected, confirmatory signature of a compound twin.

4 The representative-rotation problem

A subtlety was discovered during validation that is easy to miss: even after the branch-consistency fix above, comparing the *minimum-angle* representative of M against the 180° conditions can fail to find a genuine Type I/II relationship. This is because conjugation, $M \mapsto SMS^\top$, preserves rotation angle exactly and only rotates the axis; if the true minimum disorientation angle is far from 180° (as it generically is), no amount of searching over conjugation alone will ever produce a 180° candidate.

The complete search requires a second, non-conjugation degree of freedom, distinct from the target-side variant search introduced above: for every proper $T \in \mathcal{G}$,

$$M_T = M T \quad (6)$$

is tested against the tabulated target directions (each already representing M_{var} equivalent variants – using M_{var} here to avoid clashing with the misorientation symbol M – so that testing M_T against all M_{var} stored variants is equivalent to an additional conjugation search on the target side). Algebraically, $M_T = G_2(T^\top G_1)^\top$: this is exactly the same construction as M , but with G_1 replaced by the equally-valid symmetric branch $T^\top G_1$. The T -loop combined with the target-variant search was confirmed, by direct comparison against a brute-force search over the complete two-sided set $\{S_2 M S_1^\top : S_1, S_2 \in \mathcal{G}\}$, to reproduce the true global optimum exactly. A worked instance is given in Section 6.

5 Algorithm summary

For a candidate grain pair with (branch-consistent) average orientations G_1, G_2 :

1. Form $M = G_2 G_1^\top$.
2. For every proper symmetry operation T of the phase:
 - (a) Form $M_T = MT$ and extract its rotation axis and angle.
 - (b) For each of the three target families (n_1, a_1, n_2 ; each with M_{var} tabulated equivalent variants): find the variant minimising the combined axis and angle deviation from the condition's target axis and target angle (180° for K1_180/eta1_180, γ for K2_shear).
3. For each condition, retain the globally best (T , variant) pair found over the full search.
4. Report the pair as a match for a given condition if both the axis and angle deviations fall below a tolerance $\text{tol} = 5^\circ$ (default); report *all* conditions satisfying this, not only the single best.

For N candidate grains this is applied to every unordered pair ($\binom{N}{2}$ total), vectorised across pairs, symmetry operations, and target variants. The cost per pair scales as $O(N_s \times M_{\text{var}})$ per condition.

Relation to a naive brute-force search. A brute-force search of the complete two-sided symmetric-equivalence set against a single fixed target vector, without using the tabulated variants at all, would require looping S_1 and S_2 independently over the full point group, i.e. $O(N_s^2)$ candidate evaluations. Looping only over the M distinct tabulated variants instead of over all N_s raw symmetry operations on the target side is exact and lossless, not an approximation: by construction the M variants are precisely the distinct images produced by the N_s operations acting on the twin element tuple, so iterating the M pre-computed variants reproduces the identical set of comparisons that iterating all N_s raw operations would produce, with the redundant (duplicate-image) evaluations already removed. The T -loop-plus-variant-search cost is therefore smaller than the naive $O(N_s^2)$ brute force by a factor

$$\frac{N_s^2}{N_s M_{\text{var}}} = \frac{N_s}{M_{\text{var}}} = |\text{Stab}(K_1, \eta_1)|, \quad (7)$$

i.e. exactly the stabiliser order of the twin-element tuple. Twin systems with a small number of unique variants M_{var} (a large stabiliser) are therefore searched proportionally faster; the search is never more expensive than, and is equal to, the brute-force cost only in the worst case $M_{\text{var}} = N_s$ (trivial stabiliser).

6 Worked example: the $\{114\}$ austenite twinning mode in NiTi

The general method above is illustrated with the $\{114\}$ deformation twin mode in B2 (austenite) NiTi, generated from the tabulated primary variant

$$\eta_1 = [1\ 2\ 2], \quad K_1 = (\bar{4}\ 1\ 1), \quad (8)$$

expressed in the crystal's real-space and reciprocal-space bases respectively, converted to Cartesian unit vectors a_1, n_1 via the phase's structure matrices. The remaining $M - 1$ equivalent variants are

generated by applying every proper cubic symmetry operation ($N_s = 24$) to the tuple $(K_1, K_2, \eta_1, \eta_2)$ and retaining only genuinely distinct results – two candidates are treated as equivalent if *either* their K_1 *or* their η_1 coincide (up to sign) with an already-retained variant. This is a coarser equivalence than a single-vector orbit (which, checked directly for n_1 alone, has orbit size 3 under the 24 proper rotations, not M), consistent with the general bound $M \leq N_s$ derived above; for this system the generation algorithm yields $M = 12$, i.e. a search cost reduction of a factor $N_s/M_{\text{var}} = 24/12 = 2$ relative to the naive $O(N_s^2)$ brute-force search, per the general result above.

Branch-consistency failure, observed. On real EBSD data, one grain pair’s independently-resolved (unsymmetrized) average orientations gave a raw misorientation angle of 68.66° ; after resolving both grains to branches consistent with a shared reference, the same pair’s misorientation was 37.00° , matching the true minimum disorientation and the value visible directly on the pair’s measured pole figures. Separately, for one cluster, independent per-pixel branch resolution (rather than the single-symmetry-operation-per-cluster approach specified above) was found to split its 1408 pixels across 6 distinct branches, despite an internal pixel-to-pixel orientation spread of only $\approx 6^\circ$ once resolved consistently – confirming the branch ambiguity is a real, non-negligible effect on this class of data, not a theoretical concern only.

Representative-rotation search, observed. For a confirmed grain pair, the minimum-angle representative of M had rotation angle $\approx 37^\circ$, matching the K2_shear description (axis deviation 1.28° , angle deviation 1.83°) – but this angle is nowhere near 180° , so neither K1_180 nor eta1_180 could match via conjugation alone at this representative. The T -loop search found a specific alternative representative matching K1_180 with axis deviation 1.00° and angle deviation 0.15° – an essentially exact Type I twin relationship, confirmed independently by brute-force search over the complete two-sided symmetric-equivalence set, and consistent with the pair’s directly observed coincident $\{114\}$ pole on the measured pole figures. A second, independently checked grain pair was found to satisfy K1_180, eta1_180, *and* K2_shear simultaneously, each via its own representative rotation T – the expected signature of a compound twin, per Section 3.

7 Tolerance and interpretation

Because the search evaluates, per pair and per condition, $N_s \times M_{\text{var}}$ candidate rotations (of order several hundred), a nominal tolerance tuned for a much smaller search (e.g. the original target-variant-only search) is not automatically appropriate once the T -loop is included: the enlarged search finds the true global optimum rather than an artifact of incomplete search, but the achievable minimum deviation under *no* true relationship also shrinks as the number of candidates grows. The tolerance should therefore be calibrated against the specific dataset’s orientation-measurement noise floor (typically a few degrees for EBSD) rather than treated as a universal constant, and the reported deviation values should be inspected directly (e.g. sorted by deviation) rather than relying solely on a hard pass/fail cutoff.